

SHETH L.U.J & SIR M.V COLLEGE
Data Visualization with Power BI and Tableau

Practical No: 3

Aim: Creating Advanced Visualizations in Tableau

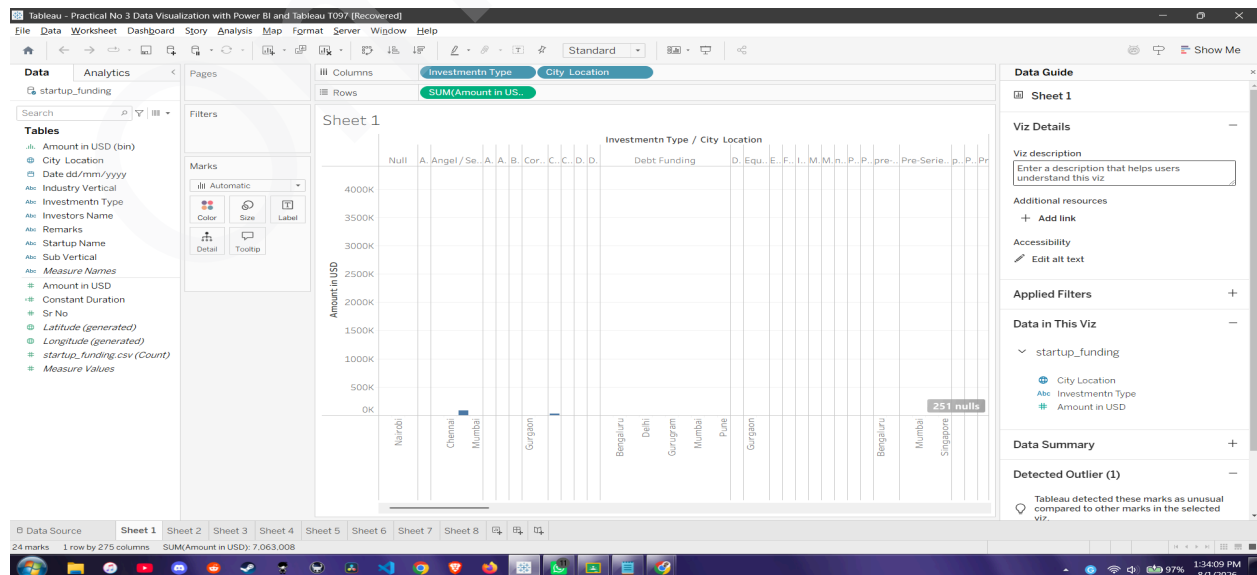
- a) Create stacked and grouped bar charts for comparison
- b) Build treemaps and pie charts for part-to-whole analysis
- c) Design scatter plots to identify relationships
- d) Create dual-axis and combination charts
- e) Develop histograms for distribution analysis
- f) Build box-and-whisker plots
- g) Use Gantt charts and advanced date visualizations

a) Create stacked and grouped bar charts for comparison

Task 1: Stacked and Grouped Bar Chart

- Create **Sheet 1**. Drag **InvestmentnType** to the **Columns** shelf and **Amount in USD** to the **Rows** shelf.
- Drag **City Location** to the **Color** tile to create a stacked bar chart.
- To convert to a grouped bar chart, drag **City Location** from the Color tile to the **Columns** shelf, placing it directly next to InvestmentnType.

After performing the following steps this is what you will be seeing:



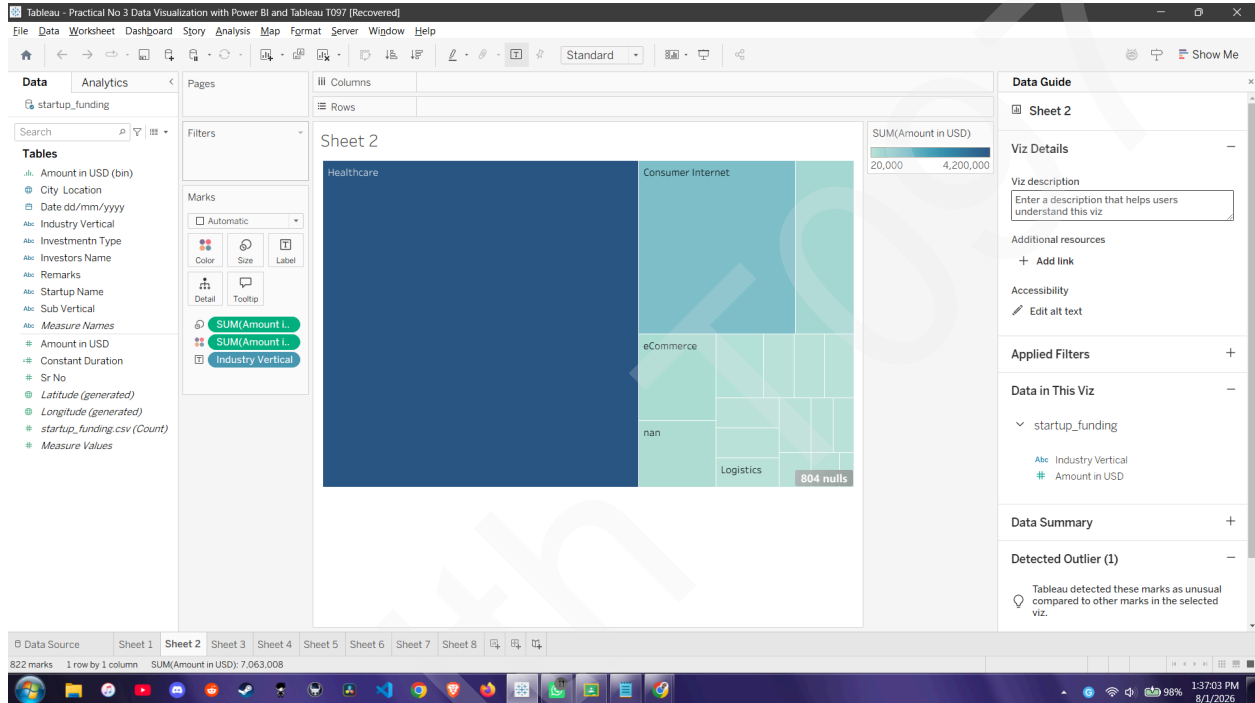
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b) Build treemaps and pie charts for part-to-whole analysis

Task 1: Treemap

- Create **Sheet 2**. Drag **Industry Vertical** to the **Columns** shelf and **Amount in USD** to the **Rows** shelf.
- Click **Show Me** $\rightarrow$ select the **Treemap** icon.

After performing the following steps this is what you will be seeing:

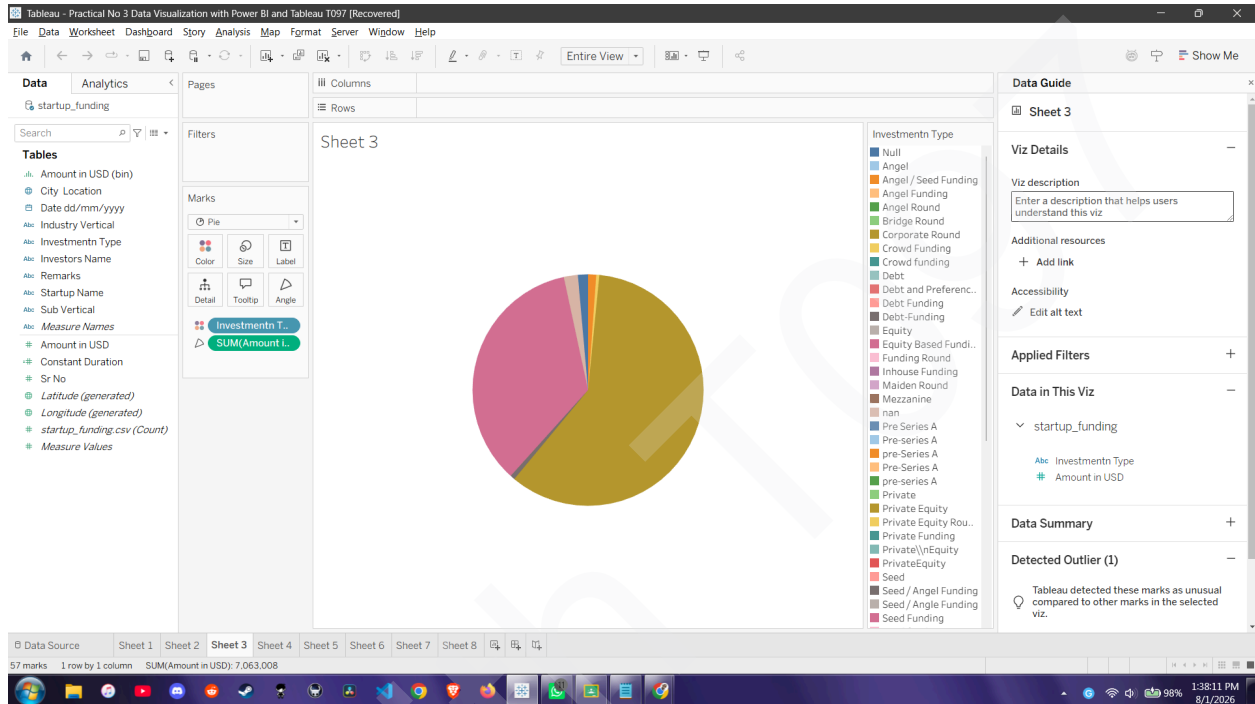


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Task 2: Pie Chart

- Create **Sheet 3**. Click **Show Me** $\rightarrow$ select the **Pie Chart** icon.
- Drag **InvestmentnType** to the **Color** tile and **Amount in USD** to the **Angle** tile.
- Set the view dropdown in the top toolbar to **Entire View** to enlarge the chart.

After performing the following steps this is what you will be seeing:



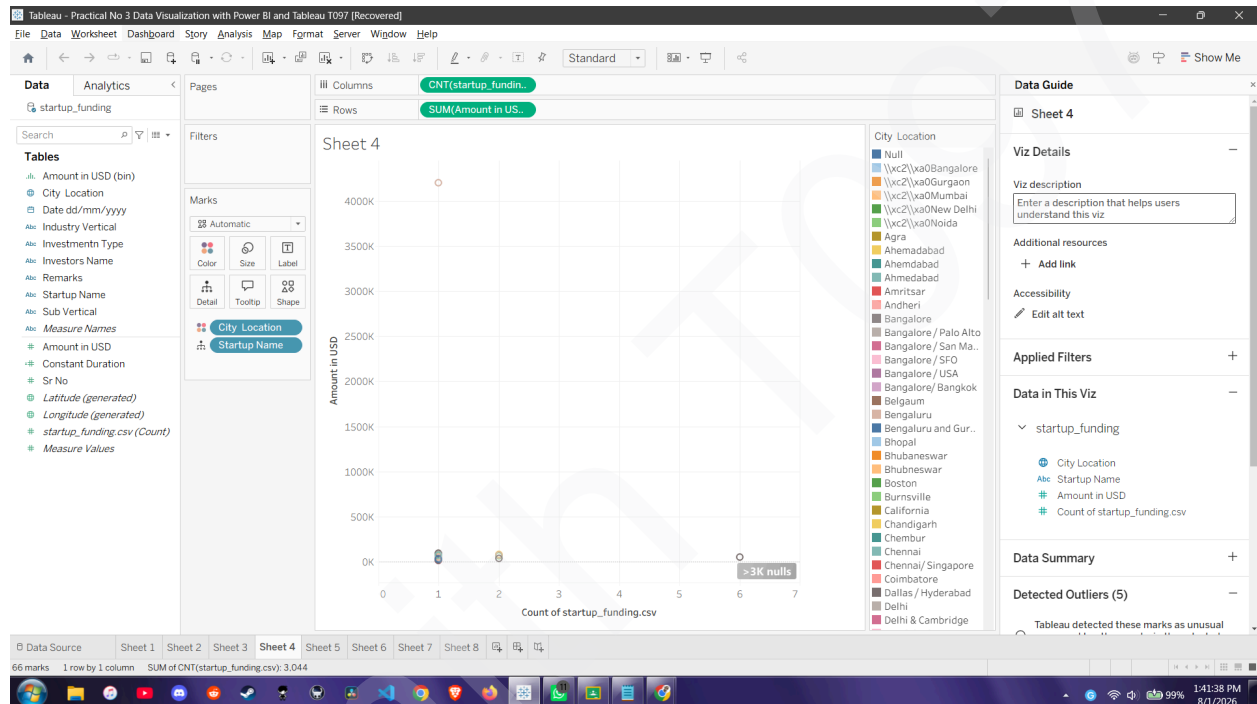
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c) Design scatter plots to identify relationships

Task 1: Scatter Plot

- Create **Sheet 4**. Drag **startup_funding.csv (Count)** to the **Columns** shelf and **Amount in USD** to the **Rows** shelf.
- Drag **Startup Name** to the **Detail** tile to disaggregate the total sum into distinct data points.
- Drag **City Location** to the **Color** tile to add visual context.

After performing the following steps this is what you will be seeing:



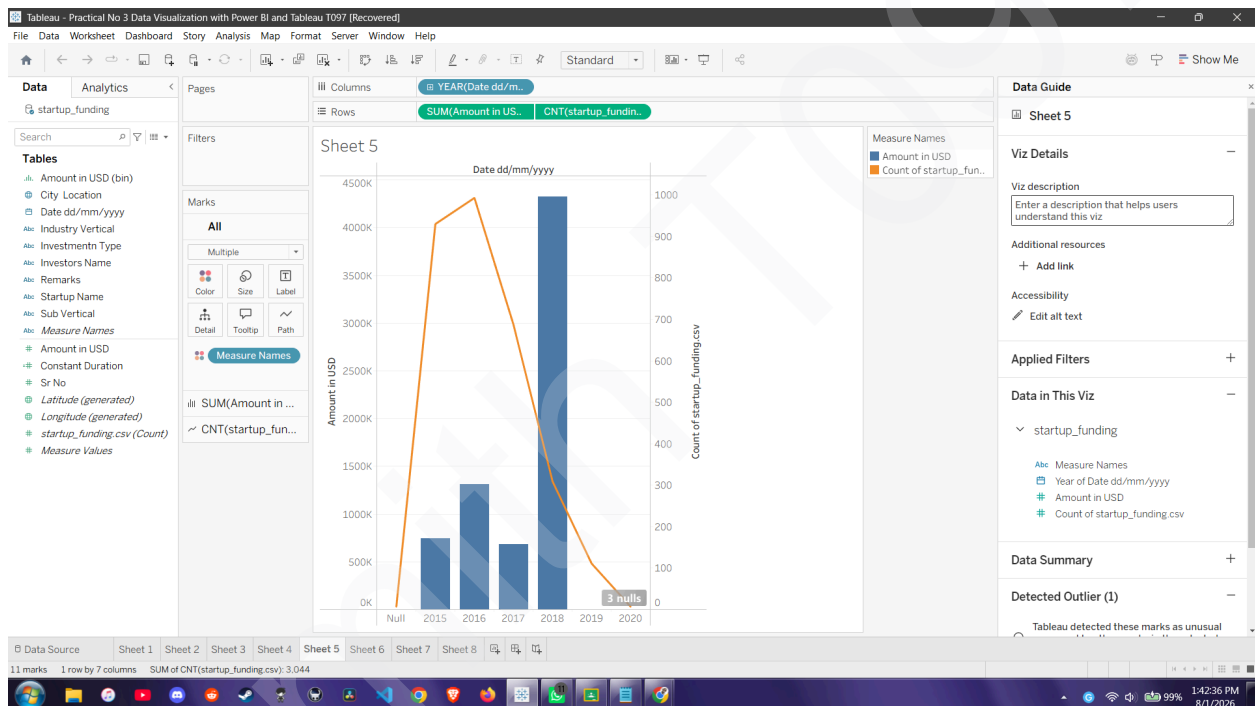
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d) Create dual-axis and combination charts

Task 1: Dual-Axis Chart

- Create **Sheet 5**. Drag **Date dd/mm/yyyy** to the **Columns** shelf and **Amount in USD** to the **Rows** shelf.
- Drag **startup_funding.csv (Count)** to the **Rows** shelf, placing it directly to the right of Amount in USD.
- Right-click the second measure pill $\rightarrow$ select **Dual Axis**.
- In the Marks card, click the SUM(Amount in USD) tab $\rightarrow$ set to **Bar**. Click the CNT(startup_funding.csv) tab $\rightarrow$ set to **Line**.

After performing the following steps this is what you will be seeing:



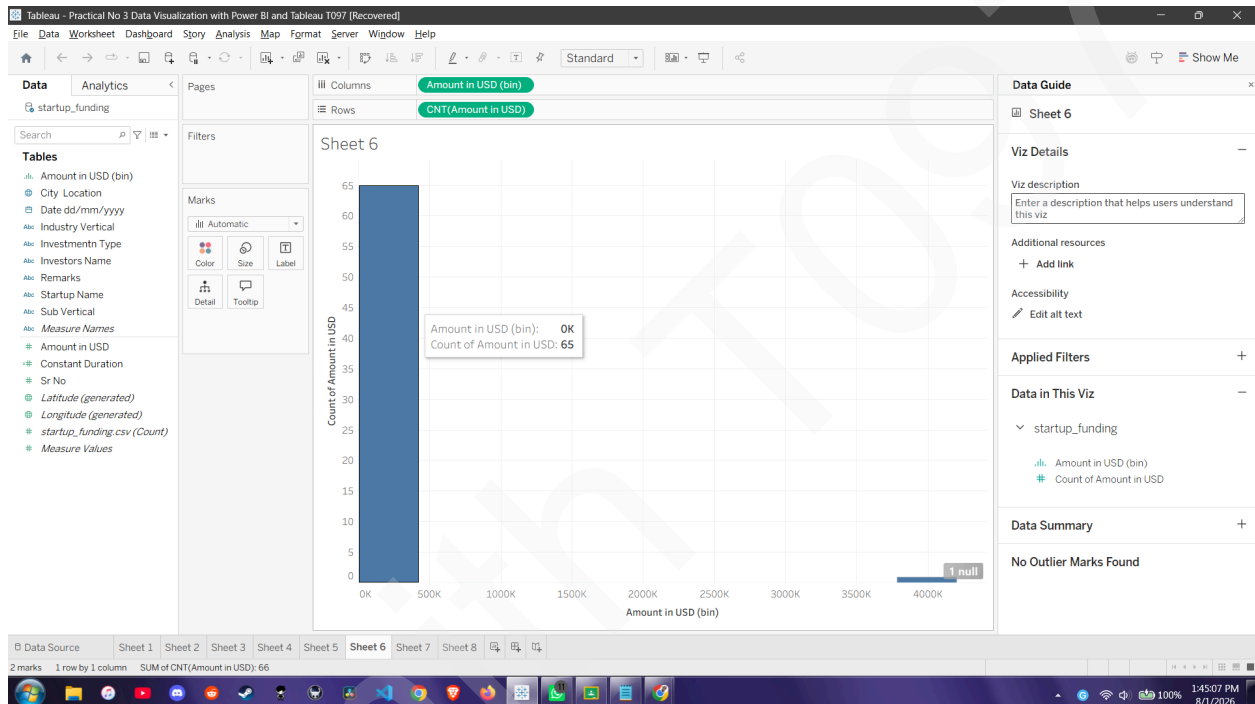
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e) Develop histograms for distribution analysis

Task 1: Histogram

- Create **Sheet 6**. Highlight **Amount in USD** in the Data pane.
- Click **Show Me** → select the **Histogram** icon.
- Tableau will automatically generate **Amount in USD (bin)** on Columns and **CNT(Amount in USD)** on Rows.

After performing the following steps this is what you will be seeing:



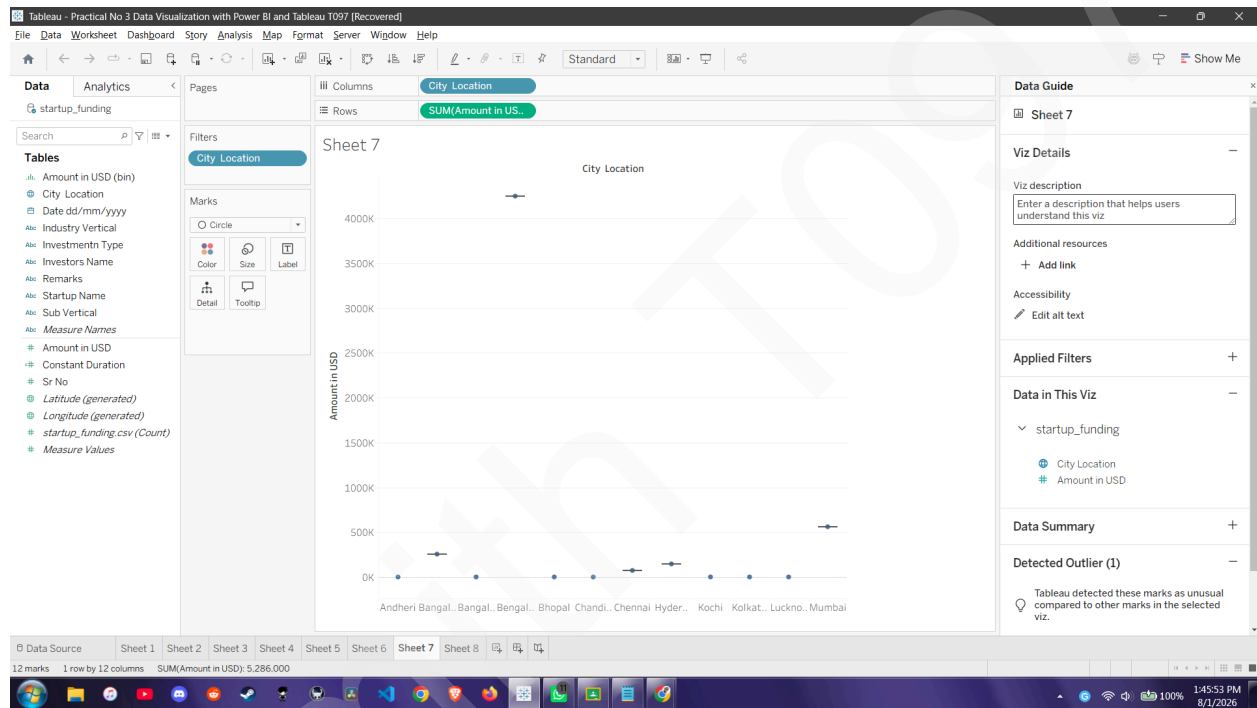
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f) Build box-and-whisker plots

Task 1: Box-and-Whisker Plot

- Create **Sheet 7**. Drag **City Location** to the **Columns** shelf and **Amount in USD** to the **Rows** shelf.
- Go to the **Analysis** menu in the top toolbar $\rightarrow$ uncheck **Aggregate Measures** to disaggregate the data points.
- Click **Show Me** $\rightarrow$ select the **Box-and-Whisker Plots** icon.

After performing the following steps this is what you will be seeing:



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g) Use Gantt charts and advanced date visualizations

Task 1: Advanced Date Gantt Timeline

Right-click the Data pane $\rightarrow$ **Create Calculated Field**. Name it Constant Duration. Formula: 1.

Create **Sheet 8**. Drag **Date dd/mm/yyyy** to the **Columns** shelf. Right-click the pill $\rightarrow$ select **Exact Date** and ensure it is set to **Continuous**.

Drag **Startup Name** to the **Rows** shelf and filter it to a specific subset of companies (e.g., Top Indian tech giants).

In the Marks card, change the mark style to **Gantt Bar**.

Drag **Constant Duration** to the **Size** tile and **Amount in USD** to the **Color** tile.

After performing the following steps this is what you will be seeing:

